

Now the key question:

Which transport should you use?

The answer is:

Use the SAME transport your legacy driver already uses.

That is overwhelmingly the safest and most correct architecture.

If legacy uses raw Ethernet

Then your Linux daemon should use:

AF_PACKET raw sockets

This is VERY common in embedded routing systems.

Architecture:

TUN ↔ userspace ↔ raw Ethernet socket

Advantages:

- preserves exact frame format
- preserves interoperability
- preserves routing behavior
- works on Zynq
- low overhead
- deterministic

This is my strongest suspicion for your system.

If legacy uses raw IP protocol numbers

Example:

IP protocol 253

then use:

SOCK_RAW

on Linux.

If legacy uses UDP encapsulation

Then use UDP sockets.

This is easiest operationally.

My recommendation for YOUR case

Based on:

- embedded routing
- legacy BSD driver
- multiple protocol instances
- node delivery
- interoperability requirement
- Zynq target

I strongly suspect your legacy system uses either:

1. raw Ethernet framing

OR

2. raw IP encapsulation

NOT UDP.

Why UDP may be a bad idea here

UDP changes semantics:

- introduces ports
- introduces kernel UDP handling
- changes MTU behavior
- changes checksum handling
- changes timing characteristics
- changes observability

Most importantly:

- legacy nodes probably won't understand it

So unless legacy already uses UDP:

do NOT introduce UDP.